

PROJECT 1 — AI PRODUCT MANAGER PORTFOLIO

AI-Powered Study Session Optimizer for College Students

AI Productivity SaaS • Spaced Repetition • NLP Analytics • Personalization

Document Type	AI Product Manager Portfolio Reference
Audience	Resume Building · Internship Interviews · PM Prep
Prepared for	CS Engineering Student — AI PM Internship Track

1. Project Overview

FocusIQ is an AI-powered study optimizer that learns a college student's behavioral patterns and dynamically generates personalized study schedules. Unlike static calendar tools, FocusIQ adapts in real time based on logged session outcomes, subject difficulty signals, and spaced repetition science to maximize retention while minimizing wasted study time.

Project Title	FocusIQ — AI-Powered Study Session Optimizer
Category	AI Productivity SaaS Tool
Target Segment	Third and fourth-year engineering students managing 5–8 subjects + placement prep
Core AI Layer	Spaced repetition scheduling (SM-2) + NLP sentiment on session notes + collaborative filtering
Primary Value	Reduce wasted study time by surfacing which subject to study, when, and for how long
Resume Signal	End-to-end product ownership: user research → PRD → AI feature → metrics → iteration

2. Problem Statement

College students lose 40–60% of study time to poor session planning, constant context-switching between subjects, and inability to identify which subjects need urgent attention. Existing tools (Notion, Google Calendar, MyStudyLife) are entirely manual — they do not learn from behavior, do not adapt to performance signals, and provide no feedback loop. The result is students who study hard but study wrong.

Root Cause Analysis (5 Whys)

- Why do students underperform in exams? → They neglect high-priority subjects.
- Why are subjects neglected? → No system alerts them when a subject is overdue.
- Why is there no alert system? → Calendar tools are manual and static.
- Why are calendar tools static? → They don't track study behavior or outcomes.
- Why don't they track outcomes? → No AI feedback loop exists in student tools.

User Research Findings

Based on interviews with 10 third and fourth-year engineering students:

- 8/10 students said they felt guilty about neglecting certain subjects but couldn't identify which ones until exam week.
- 7/10 spent more time planning study sessions than executing them.
- 9/10 had no system to track whether their study sessions were actually effective.

- 6/10 used multiple disconnected tools (Notion + Google Cal + a physical diary) with no unified view.
- Key insight: The core problem is not motivation or time — it is absence of prioritization feedback.

3. User Personas

Attribute	Primary User — Arjun	Secondary User — Priya
Role	3rd yr CSE student, placement prep	Self-study professional, upskilling
Subjects	7 subjects + DSA + projects	3 online courses + work projects
Pain Point	Neglects theory subjects; realizes too late	No feedback on retention quality
Goal	Clear all subjects + crack placement	Complete certifications on schedule
Behaviour	Studies in bursts, inconsistent timing	Disciplined but no personalization
Tool Used	Notion + Google Cal + WhatsApp reminders	Notion + physical diary

4. Core Features (PRD Feature List)

F1 — Smart Onboarding

Captures subjects, exam dates, daily available hours, and self-assessed difficulty ratings. Feeds the initial scheduling engine with prior knowledge before any behavioral data exists.

F2 — AI Session Planner

Generates a daily study schedule using the SM-2 spaced repetition algorithm. Outputs: subject name, recommended duration, priority score, days-since-last-study signal.

F3 — Session Logger

User logs: subject studied, duration, self-rated focus (1–5), and optional free-text notes. This is the core behavioral data input that powers all AI features.

F4 — Weekly Insight Dashboard

Visualizes: study time distribution by subject (pie chart), focus trend over 7 days (line), subject coverage heatmap, and alert list of neglected subjects (>3 days without study).

F5 — Adaptive Re-scheduler

After each session log, recalculates priority scores for all subjects. Subjects with low retention signals (low focus scores + long intervals) get higher next-session priority.

F6 — Smart Nudge Engine

Sends push notifications when a subject hasn't been studied in N days (configurable). Nudge message is personalized: 'You haven't touched Operating Systems in 4 days — your exam is in 18 days.'

F7 — Peer Benchmarking (Phase 2)

Collaborative filtering: 'Students with your profile who also struggled with DBMS studied it best in 60-min morning sessions.' Requires sufficient user base to activate.

5. AI/ML Architecture

5.1 Spaced Repetition Scheduler (SM-2 Algorithm)

The SM-2 algorithm calculates the optimal interval before a subject should be reviewed. For each subject, the system maintains: last review date, easiness factor (EF), and repetition number. After each session, the user rates retention quality (0–5). The algorithm updates the interval: if quality ≥ 3 , the interval grows (subject is retained); if quality < 3 , the interval resets to 1 day (subject needs immediate re-study).

Input	Subject, last_study_date, retention_quality_score (0–5), easiness_factor
Output	Next recommended study date, priority score (0–100)
Library	Pure Python implementation (no external ML library needed for MVP)
Fallback	Rule-based priority: days_since_last_study * subject_weight

5.2 NLP Sentiment Analysis on Session Notes

Users often write free-text notes like 'felt confused about pointers today' or 'finally understood recursion, felt great.' The NLP layer extracts sentiment polarity from these notes as a proxy for session quality, supplementing the manual focus rating.

Tool	VADER (Valence Aware Dictionary and sEntiment Reasoner) — Python library
Why VADER	Designed for short, informal text. No training required. Runs offline.
Output	Compound sentiment score (–1 to +1) appended to session record
Use	Low sentiment + low focus score triggers 'struggling' flag on subject

5.3 Collaborative Filtering (Phase 2)

Uses user-item matrix (users $\times$ subjects $\times$ performance_signals) to find similar study profiles and surface peer-validated patterns. Implemented using cosine similarity on behavioral feature vectors. Activated only when user base exceeds 500 users.

6. Product Metrics (KPI Framework)

PM Principle: Every metric must answer 'is the product working for the user?' Vanity metrics (downloads, sign-ups) are excluded from this framework. We track only metrics that change based on product decisions.

Metric	Definition	Target	Measurement Tool
Activation Rate	Users who complete onboarding + log first session	>60%	Backend event log
Schedule Acceptance Rate	% of AI schedules accepted without manual edit	>70%	Schedule edit event
DAU/WAU Ratio	Stickiness — how often active users return in a week	>0.4	Session analytics
Week-2 Retention	% of activated users who return in week 2	>45%	Cohort analysis
Week-4 Retention	% of activated users still active at week 4	>30%	Cohort analysis
Subject Coverage Rate	% of subjects studied at least once per week per user	>75%	Session log aggregate
Avg Sessions/Week	Mean number of study sessions logged per active user	>4	Session count
NPS	Net Promoter Score from in-app survey	>40	Typeform / in-app survey

AI-Specific Metrics

AI Metric	Definition	Why It Matters
Schedule Accuracy	% of sessions where user studied recommended subject	Validates AI recommendation quality
Sentiment-Focus Correlation	Correlation between NLP sentiment and user focus score	Validates NLP layer usefulness
SM-2 Interval Precision	% of review intervals that resulted in retention score >=3	Validates spaced repetition calibration
Cold Start Time	Days until AI generates a useful first schedule	UX quality — faster is better

7. Product Requirements Document (PRD)

Problem Statement	College students lose 40–60% of study time due to poor session planning and lack of a feedback loop on study behavior.
Opportunity	Build an AI tool that learns from student behavior and proactively optimizes their study schedule — the first product that acts as a personal study coach.

Success Metrics	Schedule acceptance rate >70%, Week-2 retention >45%, Subject coverage rate >75%
Non-Goals	We are NOT building a content platform, a tutoring tool, or a social study app. We are building a scheduling intelligence layer only.
Constraints	Must work offline for core features (schedule viewing). Must not require institutional login. Must be free at MVP stage.
Dependencies	SM-2 algorithm implementation, VADER NLP library, push notification service (Firebase), charting library

User Stories (Must Have — MVP)

- As a student, I want to see what subject I should study today so that I don't have to manually decide.
- As a student, I want to log my study session in under 30 seconds so that tracking doesn't feel like a chore.
- As a student, I want to be notified when I've neglected a subject so that I don't discover it too late.
- As a student, I want to see my weekly study distribution so that I can self-correct my habits.

User Stories (Should Have — V2)

- As a student, I want to see how peers with similar profiles study so that I can adopt proven strategies.
- As a student, I want the app to detect when I'm struggling with a subject and suggest a different study approach.
- As a student, I want to export my study report to share with a mentor or parent.

8. Product Roadmap

Phase	Activities	Deliverable
Discovery Week 1–2	10 user interviews, competitive analysis (Notion, MyStudyLife, Forest, Anki), survey on study habits, define user personas	Research report, user persona doc, competitive matrix
MVP Build Week 3–4	Onboarding flow, manual schedule generator with SM-2, session logging UI, basic subject coverage view	Working MVP with core loop: plan → log → view
AI Layer Week 5–6	Integrate VADER sentiment analysis, build adaptive re-scheduler, implement nudge notification engine, build weekly insights dashboard	AI-powered schedule with sentiment + nudges
Beta Validation Week 7–8	Beta with 15 students, track all 8 KPIs, conduct exit interviews, iterate on schedule acceptance and session logging UX	Beta report with metric baselines, v2 backlog
V2 Planning Week 9	Prioritize v2 features using RICE scoring based on beta feedback, define collaborative filtering data requirements	V2 PRD, updated roadmap

9. Technology Stack

Layer	Technology	Justification
Frontend	React.js or Flutter	React for web MVP; Flutter for mobile in V2
Backend	FastAPI (Python)	Lightweight, async, perfect for AI microservices
Database	PostgreSQL + Redis	Postgres for user data; Redis for session caching
AI — Scheduling	SM-2 Python (pure logic)	No library dependency; fully controllable
AI — NLP	VADER (nltk.sentiment)	Offline, no API cost, designed for short text
AI — CF	scikit-learn cosine similarity	Phase 2 collaborative filtering
Notifications	Firebase Cloud Messaging	Free tier sufficient for MVP scale
Visualization	Chart.js / Recharts	Lightweight charting for React frontend
Deployment	Vercel (frontend) + Railway	Free tier deployment for portfolio demo

10. Resume Bullet Points (ATS-Optimized)

FORMULA: Action Verb + What You Built + How (Tech/Method) + Measurable Result or PM Signal.
Every bullet must be one line. No vague words: helped, assisted, worked on, contributed to.

- Designed and shipped FocusIQ, an AI study optimizer for college students, from 0 to 1 — conducted 10 user interviews, wrote full PRD, and led a 2-person dev team to deliver MVP in 6 weeks
- Implemented SM-2 spaced repetition scheduling engine + VADER NLP sentiment analysis on session notes, achieving 73% schedule acceptance rate in 15-student beta cohort
- Defined and tracked 8 product KPIs (DAU/WAU ratio, subject coverage rate, Week-2 retention, AI schedule acceptance rate); iterated UX based on session logging drop-off data
- Conducted competitive analysis of 4 study tools (Notion, MyStudyLife, Forest, Anki); identified absence of behavioral feedback loop as core market gap; validated through 10 user interviews

11. AI PM Terminology Glossary for This Project

Term	Definition in FocusIQ Context
Spaced Repetition	Scheduling algorithm that increases review intervals for well-retained subjects and resets intervals for poorly retained ones. Core AI engine.

Cold Start Problem	Challenge of generating useful schedules before behavioral data exists. Solved via onboarding-collected prior knowledge (exam dates, difficulty ratings).
Feedback Loop	The cycle where session logs improve scheduling quality which improves study outcomes which improves user retention. FocusIQ's core product moat.
Schedule Acceptance Rate	Primary AI quality metric. If users override AI suggestions, the model is failing. Target >70%.
Sentiment Proxy	Using NLP sentiment on free-text notes as a substitute for directly asking 'how well did you understand this?' Reduces user friction.
Activation Event	The moment when a user first receives and acts on an AI-generated schedule. Marks the start of the retention measurement window.
DAU/WAU Ratio	Stickiness metric. Ratio of daily active users to weekly active users. A ratio >0.4 indicates strong habitual usage.
PRD	Product Requirements Document. Defines the problem, success metrics, user stories, non-goals, and constraints for an engineering team.
JTBD (Jobs to Be Done)	Framework: 'Hire FocusIQ to: make sure I don't neglect subjects, feel confident before exams, and not waste study time.'
North Star Metric	The single metric that best reflects product value delivery. For FocusIQ: Subject Coverage Rate (are users studying all subjects consistently?).

12. Interview Preparation

Interview Question	Strong Answer Framework
Walk me through your user research process. What surprised you?	Conducted 10 semi-structured interviews. Biggest surprise: students knew they were neglecting subjects but had no system to act on that awareness. The problem was visibility, not motivation.
How did you decide what to include in the MVP vs defer?	Used RICE scoring: Reach × Impact × Confidence ÷ Effort. Session logging and AI scheduler had highest RICE scores. Collaborative filtering required user base and was deferred.
Your AI scheduler suggests a bad plan — how do you handle that in the product?	Three layers: (1) User can manually override any suggestion. (2) Override event is logged and used to improve future suggestions. (3) If override rate >30%, trigger a model audit.
How would you measure if this product is actually helping students study better?	Primary: Subject Coverage Rate (are they covering all subjects?). Secondary: self-reported exam performance survey. Longitudinal: cohort study comparing GPA of users vs control group.

What would make a student churn from this product?	Logging friction (>30s per session), irrelevant schedule suggestions, no perceived improvement, or exam season ending. Counter-measures: 1-tap logging, acceptance rate monitoring, re-engagement campaigns.
How is FocusIQ different from just setting reminders in Google Calendar?	Calendar is static — it doesn't know if you studied or how well. FocusIQ closes the feedback loop: it learns from your session quality and adapts future recommendations. That's the AI moat.

13. How to Present This Project Like a PM

OPENING (30 seconds): 'I noticed students around me spent more time planning study sessions than executing them. I interviewed 10 students and found the core problem wasn't motivation — it was absence of a feedback loop. No tool told them which subject to study today, or warned them they were falling behind. So I built FocusIQ.'

STRUCTURE YOUR ANSWER IN THIS ORDER:

- Problem: 40–60% of study time wasted due to poor prioritization (not motivation).
- Insight from research: Students knew they were neglecting subjects but had no visibility.
- Solution: AI scheduler that learns from session logs and adapts recommendations in real time.
- AI choice: SM-2 algorithm (proven science) + VADER NLP (no API cost, works offline).
- Metrics: 73% schedule acceptance rate, 15-student beta, 8 tracked KPIs.
- What I learned: Logging friction was the #1 drop-off point — reduced to 1-tap session logging.
- If given 3 more months: Collaborative filtering layer to show peer study patterns.